

Nagarjun Mallesh

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SUMMARY

Full-stack Software Engineer with 4 years of experience building scalable systems, recently deepening expertise in **AI/ML** through Harvard CS50 AI. Started in enterprise infrastructure at HP (100K+ deployments), evolved into **cloud-native development, serverless, and event-driven architecture**. Strong in **TypeScript, Python, and Bash**, with a passion for building automation systems that eliminate manual workflows, turned 4-day processes into 15-minute scripts. Leverage **AI/GenAI** tools (**AWS Bedrock, Claude, Copilot**) not just as products but as development accelerators to simplify workflows.

EXPERIENCE

Systems Analyst | ML Technologies LLC, Boston, MA | Jul 2025 - Present

- Developed a full-stack application, Stolen Vehicle Recovery, a RAG system with TypeScript frontend and Python FastAPI backend using AWS Bedrock, and an emergency response orchestration platform with API Gateway, Lambda, and Step Functions, processing 1K+ daily queries.
- Built a text normalization tool, significantly reducing manual verification, utilizing Python/FastAPI for the backend, Streamlit for the frontend, and leveraging AWS Bedrock for processing. Implemented multi-threading to handle and process multiple comments efficiently.
- Provisioned all AWS infrastructure using Terraform for Infrastructure as Code, deployed and monitored services with CloudWatch.
- Automated development workflows with Bash scripts for environment setup, dependency management, and testing, reducing onboarding from 4 hours to 15 minutes, and applied GenAI tools (Claude, Copilot) for code generation, accelerating delivery.

Backend Software Engineer | Keelworks Foundation, Oak Harbor, WA | Jul 2024 - Jul 2025

- Designed a hybrid cloud storage solution using **AWS S3** and **Google Cloud Storage**, deployed infrastructure using Terraform for IaC with EventBridge-Lambda event processing, and Dockerized **Node.js** application on **AWS ECR**.
- Partnered with DBA to design a **PostgreSQL** database schema for a volunteer management system, deployed on **AWS RDS Multi-AZ** for global accessibility and automatic failover.
- Implemented CloudWatch monitoring across **Lambda, EC2, RDS, and API Gateway** with **EventBridge** rules detecting response delays and execution failures, triggering automated **SES** email alerts with error details to the engineering team.

Software Engineer | Hewlett-Packard Inc, Bangalore, India | Oct 2019 - Jul 2022

- Automated BIOS configuration deployment using **PowerShell**, transforming 4-day manual processes into 15-minute workflows and eliminating 90% of configuration.
- Orchestrated cross-functional engineering and QA teams to build a **React.js/Node.js** management dashboard for hardware provisioning.
- Led **AWS-based** OS deployment automation, reducing provisioning from days to hours, and deployed to 100K+ client machines.

Associate Software Engineer | Digital API Craft Pvt Ltd, Bangalore, India | Jul 2019 - Oct 2019

- Migrated user services module from monolith to Spring Boot microservices with Kafka messaging, handling account transactions, balance updates, and personal information access for banking system processing 50K+ daily transactions.
- Built a domain provisioning system using Golang with DNS integration, automating client onboarding.

SKILLS

Programming & Scripting: Python, JavaScript/TypeScript, Java, Bash, PowerShell

Cloud & Infrastructure: AWS (Lambda, EC2, API Gateway, CloudFormation, CloudWatch), Docker, Terraform

CI/CD & Automation: GitHub Actions, Blue-Green Deployments, Infrastructure as Code, Automated Testing

Monitoring & Reliability: CloudWatch Metrics/Alarms, Logging, Incident Response, 99.9% Uptime Delivery

Learning (SRE Transition): Kubernetes, GCP/GKE, Prometheus, Grafana, SLI/SLO Design

Data Systems: PostgreSQL, MongoDB, Redis, Kafka, DynamoDB

CERTIFICATIONS

CS50's Introduction to Artificial Intelligence with Python | Harvard University | 2024

EDUCATION

MS Information Systems | *Northeastern University, Boston, MA* | 2022-2024

Relevant Coursework: Introduction to AI, Generative AI Systems, Machine Learning

BE Information Science & Engineering | *Visvesvaraya Technological University, India* | 2015-2019

NOTABLE PROJECTS

- **Multi-Agent Travel System:** AWS Bedrock orchestration platform with specialized AI agents (Airbnb search, hotel bookings, restaurant recommendations) coordinated by a collaborator agent, using AWS Knowledge Base for property and venue information, serving 2K+ monthly users with intelligent travel planning
- **FiveAlive Virtual Cooking Assistant:** Full-stack AI application using **Gemini Vision** API to analyze fridge contents from photos, categorize ingredients (perishable, dietary preferences), generate 5-ingredient recipes with step-by-step multimodal guidance (text, audio, chat), and **OCR-based** menu allergen detection with internet verification